

Task1: Process Synchronization and Child Process Monitoring Using waitpid()

```
printf("Parent process started\n");
printf("Parent PID: %d\n", getpid());

int status;
pid_t result = waitpid(pid, &status, 0);

if (result == -1) {
    perror("waitpid");
    return 1;
}

if (WIFEXITED(status)) {
    printf("Child exited normally\n");
    printf("Exit status: %d\n", WEXITSTATUS(status));
}

return 0;
}
```

GNU nano 7.2

waitpid_test

```
#include <stdio.h>
#include <unistd.h>
#include <sys/wait.h>
#include <stdlib.h>

int main() {
    pid_t pid = fork();

    if (pid < 0) {
        perror("fork");
        return 1;
    }

    if (pid == 0) {
        printf("Child process started\n");
        printf("Child PID: %d\n", getpid());

        sleep(2);

        printf("Child process finished\n");
        exit(42);
    }

    printf("Parent process started\n");
    printf("Parent PID: %d\n", getpid());
}
```

```

shreyas@Grigori:~$ mkdir -p ~/skill_session_7
shreyas@Grigori:~$ cd~/skill_session_7
-bash: cd~/skill_session_7: No such file or directory
shreyas@Grigori:~$ cd ~/skill_session_7
shreyas@Grigori:~/skill_session_7$ nano waitpid_test.c
shreyas@Grigori:~/skill_session_7$ gcc waitpid_test.c -o waitpid_test
shreyas@Grigori:~/skill_session_7$ ./waitpid_test
Parent process started
Child process started
Child PID: 488
Parent PID: 487
Child process finished
Child exited normally
Exit status: 42
shreyas@Grigori:~/skill_session_7$ ps
  PID TTY          TIME CMD
  459 pts/0        00:00:00 bash
  489 pts/0        00:00:00 ps
shreyas@Grigori:~/skill_session_7$ |

```

Task 2: To Retrieve PATH Variable, Parse Search Directories, Locate Executables, Verify Permissions, Handle Missing Commands, Test Command Resolution

```

shreyas@Grigori:~/skill_session_7$ nano path_search.c
shreyas@Grigori:~/skill_session_7$ gcc path_search.c -o path_search
shreyas@Grigori:~/skill_session_7$ echo $PATH
/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/usr/games:/usr/local/games:/usr/lib/wsl/lib:/mnt/c/Python314/Scripts:/mnt/c/Python314:/mnt/c/Program Files/Common Files/Oracle/Java/javapath:/mnt/c/Program Files (x86)/Common Files/Oracle/Java/javapath:/mnt/c/Program Files (x86)/Common Files/Oracle/Java/javapath:/mnt/c/Windows/system32:/mnt/c/Windows:/mnt/c/Windows/System32/Wbem:/mnt/c/Windows/System32/WindowsPowerShell/v1.0:/mnt/c/Windows/System32/OpenSSH:/mnt/c/Program Files (x86)/NVIDIA Corporation/PhysX/Common:/mnt/c/Users/shrey/Downloads/jdk-25_windows-x64_bin/jdk-25/bin:/mnt/c/Program Files/NVIDIA Corporation/NVIDIA App/NvDLISR:/mnt/c/Program Files/Git/cmd:/mnt/c/Program Files/nodejs:/mnt/c/ProgramData/chocolatey/bin:/mnt/c/Program Files/PuTTY:/mnt/c/Program Files/MySQL/MySQL Shell 8.0/bin:/mnt/c/Users/shrey/.local/bin:/mnt/c/Users/shrey/Downloads/flutter_windows_3.35.6-stable/flutter/bin:/mnt/c/Users/shrey/AppData/Local/Programs/Python/Python313/Scripts:/mnt/c/Users/shrey/AppData/Local/Programs/Python/Python313:/mnt/c/Users/shrey/AppData/Local/Programs/Python/Launcher:/mnt/c/Users/shrey/AppData/Local/Microsoft/WindowsApps:/mnt/c/msys64/ucrt64/bin:/mnt/c/Program Files/JetBrains/CLion 2025.2/bin:/mnt/c/Users/shrey/AppData/Local/Programs/Microsoft VS Code/bin:/mnt/c/Program Files/JetBrains/IntelliJ IDEA Community Edition 2025.2.3/bin:/mnt/c/Users/shrey/AppData/Local/JetBrains/toolbox/scripts:/mnt/c/Program Files/JetBrains/WebStorm 2025.3.2/bin:/mnt/c/Users/shrey/AppData/Local/Programs/Antigravity/bin:/mnt/c/Program Files/JetBrains/PyCharm 2025.2.4/bin:/mnt/c/Users/shrey/AppData/Roaming/npm:/mnt/c/Users/shrey/AppData/Local/Python/bin:/mnt/c/Users/shrey/AppData/Local/GitHubDesktop/bin:/mnt/c/Users/shrey/AppData/Local/Programs/Antigravity IDE/bin:/mnt/c/Users/shrey/Downloads/pypy3.12-v8.0.0-win64:/snap/bin
shreyas@Grigori:~/skill_session_7$ ./path_search ls
Executable found: /usr/bin/ls
shreyas@Grigori:~/skill_session_7$ ./path_search gcc
Executable found: /usr/bin/gcc
shreyas@Grigori:~/skill_session_7$ ./path_search bash
Executable found: /usr/bin/bash
shreyas@Grigori:~/skill_session_7$ ./path_search abcdxyz
Command not found: abcdxyz
shreyas@Grigori:~/skill_session_7$ ./path_search ls
./path_search pwd
./path_search echo

```

```
shreyas@Grigori:~/skill_session_7$ ./path_search ls
./path_search pwd
./path_search echo
./path_search cat
./path_search nonexistentcommand
Executable found: /usr/bin/ls
Executable found: /usr/bin/pwd
Executable found: /usr/bin/echo
Executable found: /usr/bin/cat
Command not found: nonexistentcommand
shreyas@Grigori:~/skill_session_7$ |
```

```
GNU nano 7.2 path_search.c *
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <unistd.h>

int main(int argc, char *argv[]) {
    if (argc != 2) {
        printf("Usage: %s <command>\n", argv[0]);
        return 1;
    }

    char *path = getenv("PATH");

    if (path == NULL) {
        printf("PATH variable not found\n");
        return 1;
    }

    char *path_copy = strdup(path);
    char *dir = strtok(path_copy, ":");

    while (dir != NULL) {
        char full_path[1024];
```

```
while (dir != NULL) {  
    char full_path[1024];  
    snprintf(full_path, sizeof(full_path), "%s/%s", dir, argv[1]);  
    if (access(full_path, X_OK) == 0) {  
        printf("Executable found: %s\n", full_path);  
        free(path_copy);  
        return 0;  
    }  
    dir = strtok(NULL, ":");  
}  
printf("Command not found: %s\n", argv[1]);  
free(path_copy);  
return 1;  
}
```